

ELIAS ANTOUN

Computer Engineer

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OBJECTIVE

Fourth-year Computer and Communications Engineering student with a perfect GPA and Dean's Honor List recognition. Experienced in Python, AI/ML, ROS2-based navigation stacks, containerized applications, and AI-driven automation pipelines. Actively engaged in projects that turn engineering concepts into working prototypes across software and hardware.

EDUCATION

Notre Dame University - Louaize (NDU) , Zouk Mosbeh, Lebanon	2023 - 2027
BE in Computer and Communication Engineering	
- Cumulative GPA: 4.0/4.0 - Awards: Dean's Honor List - every semester (Fall 2023 - Spring 2026)	

WORK EXPERIENCE

Teaching and Lab Assistant - Notre Dame University - Louaize (NDU)	Sept 2024 - Present
- Supporting students in circuit analysis, simulations, and hardware implementation. - Developed tutorials for VHDL simulation using Synopsys tools to support digital design coursework.	
Technical Coordinator - IEEE NDU MC	Sept 2024 - Present
Organizing and coordinating technical workshops and engineering events for IEEE student members.	
Private Tutor	Sept 2023 - Present
Assisting grade 9 to 12 students and university students in math, physics, chemistry.	
Robotics Intern - inmind.academy, Tal El Zaatar, Lebanon	Jun 2026 - Aug 2026
- Built and containerized ROS2 applications with CMake, Docker, and Docker Compose. - Implemented SLAM, path planning, and computer vision pipelines for autonomous robot navigation.	
AI Engineer Intern - ANB / Layer10, Nahr el Mot, Lebanon	May 2026 - Aug 2026
- Replaced manual accounting workflows at a retail automotive group with autonomous document-processing agents, handling 1,000 invoices/month and cutting manual review time by 90%. - Raised document classification accuracy from 81.4% to 95.3% through evolutionary prompt optimization on a 165-PDF labeled set. - Developed knowledge graph embeddings and visual similarity models powering supplier resolution and product matching. - Benchmarked agent harnesses on token cost and extraction accuracy, designing a leakage-controlled evaluation cohort via UMAP/HDBSCAN clustering over 12,375 documents.	

PROJECTS

Hand Gesture Controlled Robotic Car – Python, Computer Vision, Embedded Control
• Developed an interactive motor control system using OpenCV and Mediapipe. • Implemented real-time gesture recognition for wireless control via NodeMCU. • Built a Python GUI for live monitoring and user interaction.
License Plate Recognition System – MATLAB, Image Processing, Pattern Recognition
• Designed a detection and recognition system using MATLAB App Designer. • Applied FFT encryption/decryption, edge detection, and correlation techniques. • Implemented automated character extraction and recognition.

CERTIFICATIONS & TRAINING

IP Design Training Program - SiliconCedars / Synopsys	Aug 2025
IC fundamentals, fabrication, and packaging. Sequential circuits, counters, registers, RAM/ROM design.	
Cisco Networking Academy - NetAcad Summer Camp	Aug 2024
- Network Technician Career Path. - Python Essentials.	

TECHNICAL SKILLS

Programming: Python, C++, C#, Bash, SQL AI/ML: Claude Agent SDK, AWS Bedrock, Neo4j, RAG & embeddings, UMAP/HDBSCAN, PyTorch Infrastructure: Docker, AWS (S3, SQS, Lambda), Oracle, PostgreSQL, Grafana, Git Embedded & Hardware: Arduino, ESP32, NodeMCU, VHDL, MATLAB/Simulink, Cadence Virtuoso		
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SOFT SKILLS

- Teamwork - Effective Communication - Adaptability

LANGUAGES

- English: C2 - French: C1 - Arabic: Native

INTERESTS

- Radio communication - Satellites and space systems - Physics
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